

Mohammed Zainul Abedin Kaleemi

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EDUCATION

Muffakham Jah College of Engineering & Technology (MJCET)

Expected May 2028

B.E. in Mechanical Engineering, Osmania University

- **Coursework:** Scientific computing (Python, C, MATLAB), system modeling, mechanical system analysis, thermodynamics and materials science labs, embedded systems (AVR/8051), statistics, CAD (SolidWorks, AutoCAD)

EXPERIENCE

Team Robocon MJCET

Nov 2024 – Present

Mechanical Head (promoted from Design Engineer and Senior Design Engineer)

- Lead the mechanical team for the ABU Robocon robot, from design and prototyping to fabrication and assembly.
- Design robot mechanisms for kinematics, strength, weight and reliability, iterating through testing.
- Produce 3D CAD models and drawings; coordinate integration with the electrical and programming teams.

Team MudBrothers (MJCET)

Mar 2026 – Present

Vehicle Design Engineer, SAE BAJA

- Design parts and assemblies in CAD for the team's off-road vehicle.
- Work on subsystem design and integration, balancing weight, reliability and what can be fabricated.
- Work with the manufacturing and testing teams to revise designs; help with fabrication, assembly and testing.

Jawaharlal Nehru Technological University Hyderabad (JNTUH)

Jul – Aug 2026

EV Design, Development & Manufacturing Intern

- Hybrid internship covering EV architecture, battery packs, Battery Management Systems (BMS), charging systems, MATLAB/Simulink and EV digital twins.
- Attended industry visits and expert sessions on EV development and manufacturing.

PROJECTS

S.A.F.L. (Smart Agri Four-Legged Bot)

Nov 2025 – Present

Undergraduate Researcher, institution-funded R&D program

- Contributed to the mechanical design of a semi-autonomous agricultural quadruped: 3D CAD assemblies, reinforced steel chassis and articulated legs.
- Performed FEA in SolidWorks on critical components; structure validated to a minimum safety factor of 2.04.
- Designed mechanical interfaces for the actuators, soil sensing, cameras and onboard computer; assisted with Jetson Nano edge AI and sensor acquisition.
- Built and tested the prototype in field trials, including 4–6 hour field deployments.

Wearable Exoskeleton System

Nov 2025 – Present

R&D project, supported under the YUKTI Innovation Challenge

- Wearable assist system to reduce strain and fatigue in manual tasks.
- Worked on the mechanical integration and assembly of a rope-driven assist using Dyneema cable, high-torque servos and 3D-printed parts.
- Controlled the servos from a headless Raspberry Pi with a Bus Servo Driver HAT.

SKILLS

CAD & Analysis: SolidWorks (Certified Professional, CSWP), AutoCAD, FEA, Ansys Workbench

Fabrication & Systems: 3D printing (PLA/PETG), mechanical assembly, rapid prototyping, CNC, Linux (Arch)

Beginner level: Git, Jetson Nano, RealSense D435, Raspberry Pi, Arduino, servo control

Software & Media: Claude Code, vibe coding, DaVinci Resolve, Adobe Premiere Pro

AWARDS

1st Place, AgriTech track, MAKEFORHYDERABAD (Titan Design Impact Movement)

May 2026

All India Rank 10, IKR Go-Kart Competition (Team Asphalt MJCET)

LEADERSHIP

Club Optimus MJCET, Technical Head

Sep 2026 – Present

IEOM MJCET, Social Media Director

Sep 2026 – Present